

- 29 A metal surface is illuminated with a beam of monochromatic electromagnetic radiation. By the photoelectric effect, photoelectrons may be emitted from the metal surface.

Which statement about the photoelectrons is correct?

- A No emission of photoelectrons occurs if the radiation is of a very low intensity.
- B Photoelectrons are emitted only if the wavelength of the radiation is greater than a minimum value.
- C The maximum speed of the photoelectrons emitted increases when the intensity of the radiation increases at constant frequency.
- D The rate of emission of photoelectrons decreases when the frequency of the radiation increases at constant intensity.

Ans: D

Option A is incorrect because photoelectric effect can occur even for very low intensity but sufficiently high frequency radiation (above the threshold frequency f_0).

Option B is incorrect because photoelectrons are emitted only if the frequency of radiation is greater than a minimum frequency known as threshold frequency f_0 ; since $\lambda_0 = \frac{c}{f_0}$, the wavelength of the radiation must be smaller than the threshold wavelength, which is instead a maximum value.

Option C is incorrect because the maximum kinetic energy and thus speed of the photoelectrons is independent of intensity but depends on the frequency of radiation and work function of the metal, as given by $hf = \Phi + E_{k,max}$.

Option D is correct since $I = \frac{N_P hf}{t A}$, at constant intensity and increased frequency, the rate of photons incident on the metal decreases, the lesser the rate of emission of photoelectrons.

30 The line emission spectrum of a hydrogen discharge tube when viewed through a diffraction